State-chart Diagram Guide

4/2019

What is state-chart diagram

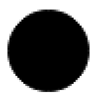
- define different states of an object during its lifetime and these states are changed by events.

Basic components of a statechart diagram

1. Initial state
2. Transition
3. State
4. Self transition
5. Composite state
6. Final state

Basic components

1. Initial state – We use a black filled circle represent the initial state of a System or a class.

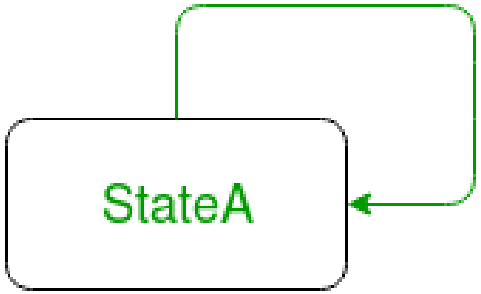


2. Transition – We use a solid arrow to represent the transition or change of control from one state to another. The arrow is labelled with the event which causes the change in state

Basic components

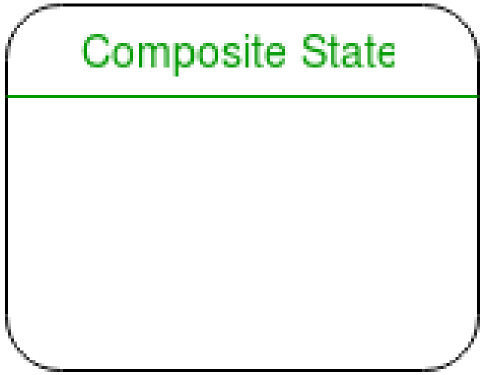
3. State – We use a rounded rectangle to represent a state. A state represents the conditions or circumstances of an object of a class at an instant of time.

4. Self transition – We use a solid arrow pointing back to the state itself to represent a self transition. There might be scenarios when the state of the object does not change upon the occurrence of an event. We use self transitions to represent such cases.

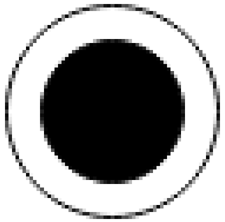


Basic components

5. Composite state – We use a rounded rectangle to represent a composite state also.We represent a state with internal activities using a composite state.



6. Final state – We use a filled circle within a circle notation to represent the final state in a state machine diagram.



Basic components

Join – We use a rounded solid rectangular bar to represent a Join notation with incoming arrows from the joining states and outgoing arrow towards the common goal state. We use the join notation when two or more states concurrently converge into one on the occurrence of an event or events. ????

Fork – We use a rounded solid rectangular bar to represent a Fork notation with incomig arrow from the parent state and outgoing arrows towards the newly created states. We use the fork notation to represent a state splitting into two or more concurrent states.

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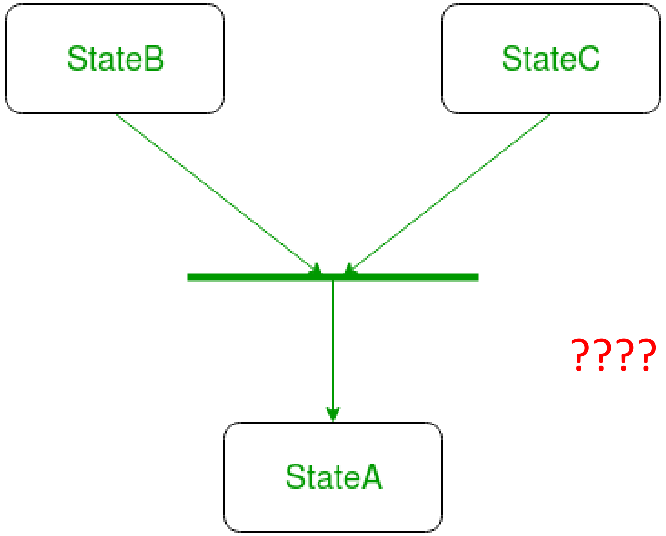

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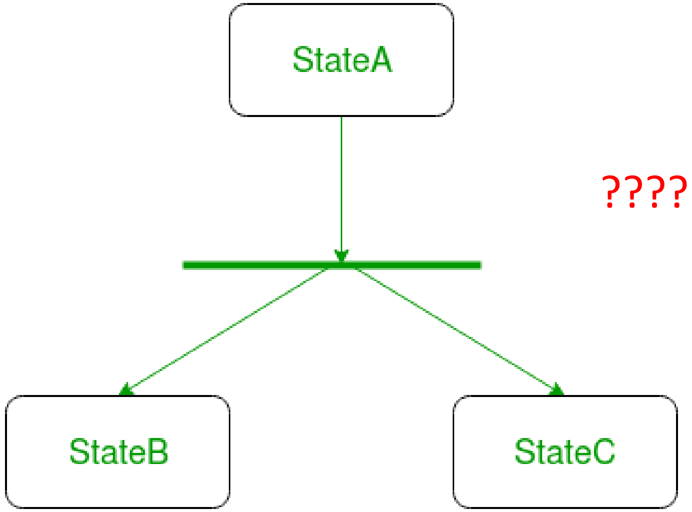

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Steps to draw a state diagram

- Identify the initial state and the final terminating states.
- Identify the possible states in which the object can exist (boundary values corresponding to different attributes guide us in identifying different states).
- Label the events which trigger these transitions.

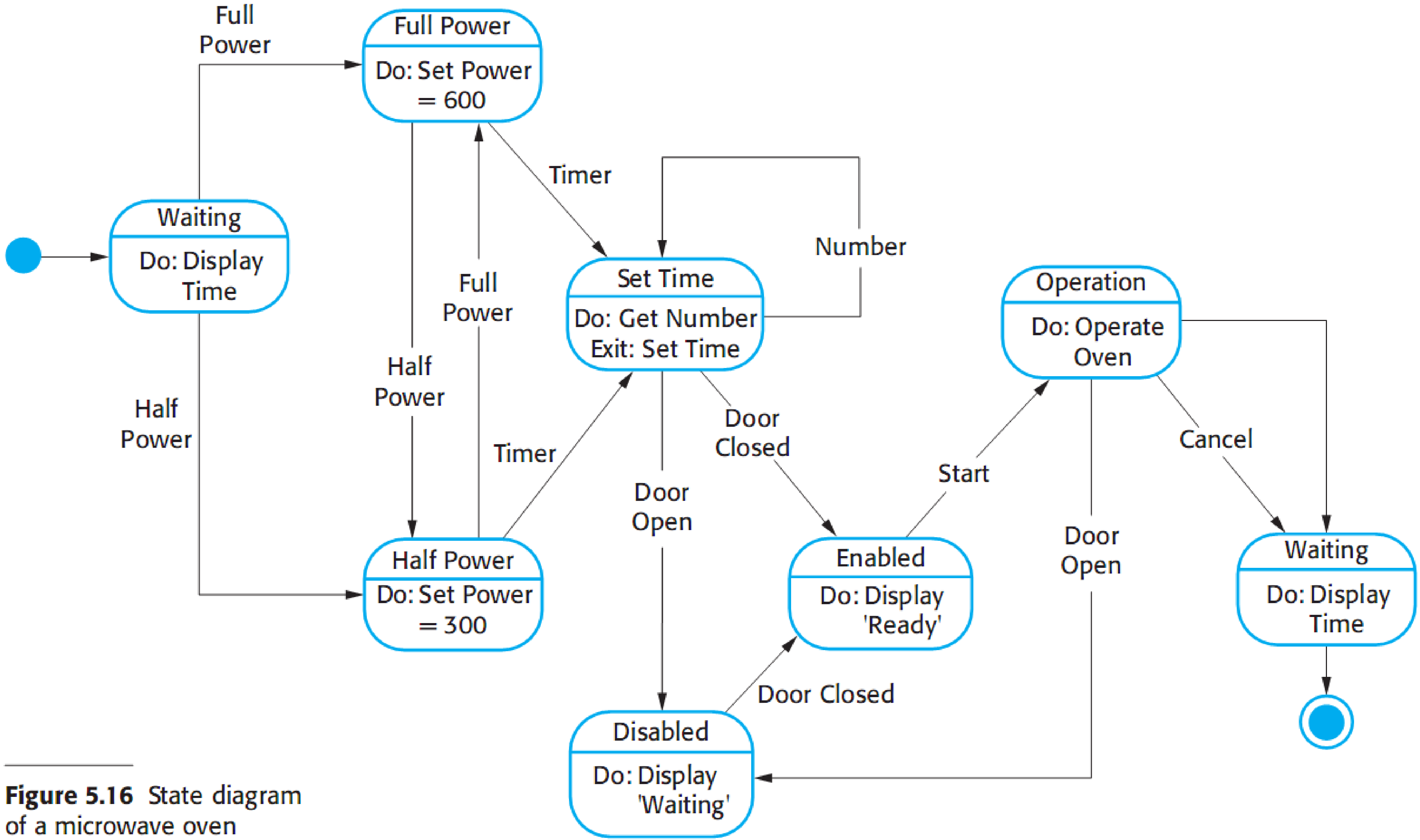

Figure 5.16 State diagram of a microwave oven

State	Description
Waiting	The oven is waiting for input. The display shows the current time.
Half power	The oven power is set to 300 watts. The display shows 'Half power'.
Full power	The oven power is set to 600 watts. The display shows 'Full power'.
Set time	The cooking time is set to the user's input value. The display shows the cooking time selected and is updated as the time is set.
Disabled	Oven operation is disabled for safety. Interior oven light is on. Display shows 'Not ready'.
Enabled	Oven operation is enabled. Interior oven light is off. Display shows 'Ready to cook'.
Operation	Oven in operation. Interior oven light is on. Display shows the timer countdown. On completion of cooking, the buzzer is sounded for five seconds. Oven light is on. Display shows 'Cooking complete' while buzzer is sounding.

Stimulus	Description
Half power	The user has pressed the half-power button.
Full power	The user has pressed the full-power button.
Timer	The user has pressed one of the timer buttons.
Number	The user has pressed a numeric key.
Door open	The oven door switch is not closed.
Door closed	The oven door switch is closed.
Start	The user has pressed the Start button.
Cancel	The user has pressed the Cancel button.

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1. Diagram
2. Tabular of State desc
3. Tabular of Stimulus desc

Note [1]

- Đủ start và final
- Cần định nghĩa event
  - “Xác nhận xuất hàng”, “Nhiệt độ vượt ngưỡng”, ...
- Tất cả các chuyển trạng thái (mũi tên) phải có event
- Tên các state là danh từ/danh động từ chỉ trạng thái
  - “Đang chờ”, “Hết hàng”, ...
  - Không trùng nhau!
- Sử dụng action trong mỗi state
  - enter: thực hiện action khi bắt đầu vào trạng thái
  - exit: thực hiện action ngay trước khi kết thúc trạng thái
  - do: thực hiện action khi vẫn còn trạng thái

Note [2]

- Không phải là “screen flow” / “menu chart” !